

15 - Halogen compounds

30/12/24

Very few halogenoalkanes are found in nature, most are made in chemical reactions with a halogen or halide compound

Production of halogenoalkanes

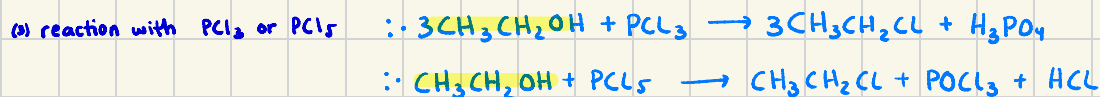
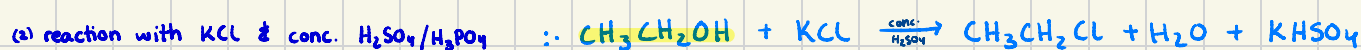
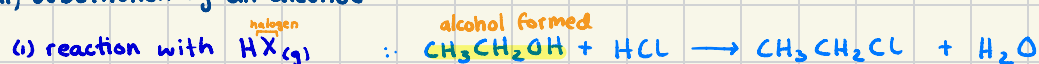
i) free-radical substitution of alkanes by Cl_2 or Br_2 in UV light



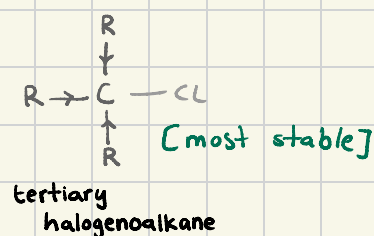
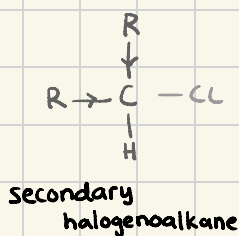
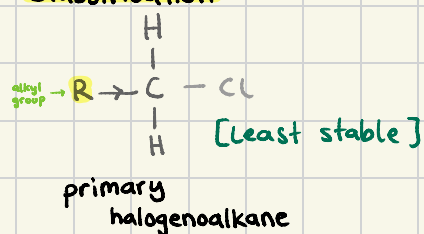
ii) electrophilic addition of an alkene with a halogen or halide compound at room temp.



iii) substitution of an alcohol

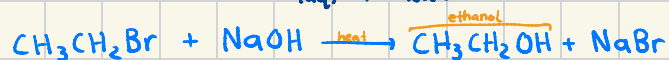


Classification



Substitution reactions of halogenoalkanes

i) reaction with $\text{NaOH}_{(\text{aq})}$ & heat



if we use $\text{NaOH}_{(\text{aq})}$, a nucleophilic substitution takes place.

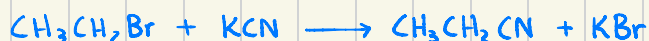
if we use $\text{NaOH}_{(\text{ethanol})}$, an elimination reaction takes place

* elimination reaction with NaOH in ethanol & heat to produce an alkene

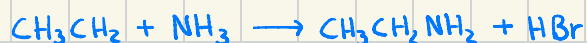


OH^- from NaOH accepts a hydrogen atom (forming water) & the $\text{C}-\text{Br}$ bond breaks heterolytically (forming Br^- with Na)

ii) reaction with KCN & heat



iii) reaction with NH_3 in ethanol heated under pressure



iv) reaction with aqueous silver nitrate in ethanol & heat

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- the halide in the halogenoalkane combines with a silver ion to form a silver halide ppt.

- the rate of formation of the ppt. can be used to compare the reactivity of the different halogenoalkanes

* quicker the ppt. forms, faster the substitution & more reactive the halogenoalkane

AgCl - white ppt.

AgBr - cream ppt.

AgI - yellow ppt.

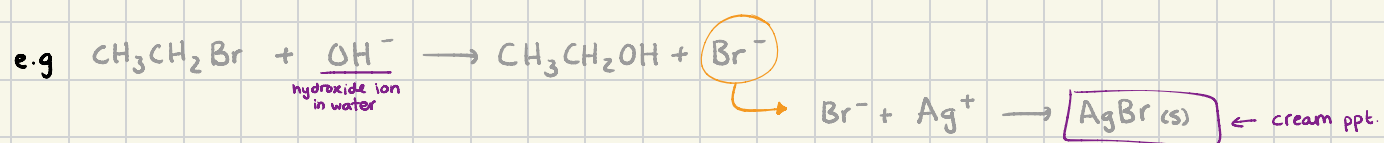
quicker formation
quicker substitution
more reactive

→ fastest nucleophilic substitution → iodoalkanes

slowest nucleophilic substitution → fluoroalkanes

iodoalkanes are the most reactive halogenoalkanes $\because$ C-I bond is the weakest

fluoroalkanes are the least reactive halogenoalkanes $\because$ C-F bond is the strongest



Nucleophilic substitution reactions

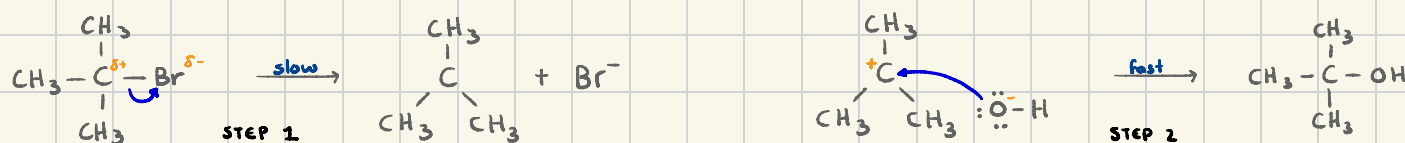
1) unimolecular nucleophilic substitution → $\text{S}_{\text{N}}1$ mechanism

2) bimolecular nucleophilic substitution → $\text{S}_{\text{N}}2$ mechanism

$\text{S}_{\text{N}}1$ mechanism

- tertiary halogenoalkanes tend to react via $\text{S}_{\text{N}}1$ mechanism

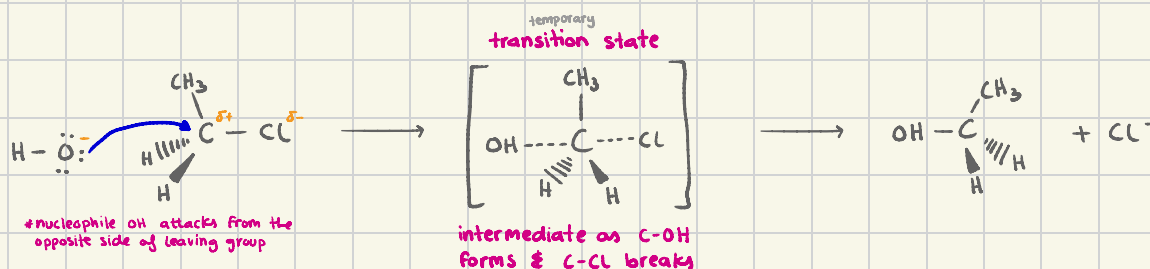
- two step process



$\text{S}_{\text{N}}2$ mechanism

- primary halogenoalkanes tend to react via $\text{S}_{\text{N}}2$ mechanism

- one step process



* secondary halogenoalkanes tend to react with a mixture of $\text{S}_{\text{N}}1$ & $\text{S}_{\text{N}}2$ mechanisms